



GNC Engineer (Starfall)

Hawthorne, CA

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SpaceX was founded under the belief that a future where humanity is out exploring the stars is fundamentally more exciting than one where we are not. Today SpaceX is actively developing the technologies to make this possible, with the ultimate goal of enabling human life on Mars.

GNC ENGINEER (STARFALL)

SpaceX is looking for engineers with backgrounds in orbital mechanics, control theory, state estimation, spacecraft systems design, atmospheric entry, optimization, and software development to join the Starfall guidance, navigation, and control (GNC) team. Come help us develop the GNC algorithms for Starfall, an entry capsule designed for logistics and in-space manufacturing applications.

RESPONSIBILITIES

- Develop highly reliable and performant GNC algorithms, simulations, tools, services, and dashboards using C++ or Python; Starfall GNC Engineers write actual production code, not just prototypes for transcription by another team
- Participate in architecture, design, and code reviews
- Bulk data analysis to review performance data from Starfall flights, simulations, and operations
- Write high quality technical documentation of programs, algorithms, or analysis
- Entry vehicle trajectory design, guidance, and optimization
- Atmospheric entry, descent, and landing (EDL) algorithms
- Precision navigation and state estimation for reentry and recovery
- Attitude determination and control during orbital, deorbit, and atmospheric flight phases
- Fleet-level or multi-vehicle coordination for logistics delivery and recovery operations
- Simulation, analysis, and performance evaluation of entry systems under a wide range of conditions
- Integration of GNC with vehicle systems supporting military logistics and in-space manufacturing missions

BASIC QUALIFICATIONS

- Bachelor's degree in computer science, aerospace, physics or an engineering discipline
- 1+ years of software development experience in either C++ or Python

PREFERRED SKILLS AND EXPERIENCE

- Master's degree or PhD in an engineering discipline or physics
- Developed, debugged and deployed software that has been used in real world applications/projects
- Creative approach to problem solving, exceptional analytical skills and engineering fundamentals
- Experience with atmospheric entry guidance, navigation, and control for capsules or reentry vehicles
- Strong understanding of orbital mechanics, deorbit planning, and entry trajectory optimization
- Experience in verifying GNC algorithm performance via simulation, and validation of the simulation's models (including atmospheric, aerodynamic, and thermal environments)
- Experience with state estimation, particularly for reentry vehicles (IMU, GNSS, radar/optical, or other sensors)
- Familiarity with hypersonic flight regimes, aeroheating considerations, and precision landing/recovery techniques
- General familiarity with the disciplines necessary for spacecraft and entry vehicle flight including structures, propulsion, avionics, GNC, flight software, dynamics, launch and mission operations, and integration. Practical experience in real world development of these systems is highly desired
- Capable of identifying and solving complex problems with little to no supervision or direction
- Passion for advancing the commercial space industry and enabling new applications such as military logistics and in-space manufacturing
- Ability to work in a fast-paced, autonomously driven, and demanding start-up atmosphere
- Excellent communication skills both written and verbal

ADDITIONAL REQUIREMENTS

- Must be willing to work extended hours and weekends as needed

COMPENSATION AND BENEFITS:

Pay range:

Level I: \$125,000.00 - \$150,000.00/per year

Level II: \$145,000.00 - \$175,000.00/per year

Your actual level and base salary will be determined on a case-by-case basis and may vary based on the following considerations: job-related knowledge and skills, education, and experience.

Base salary is just one part of your total rewards package at SpaceX. You may also be eligible for long-term incentives, in the form of company stock or long-term cash awards, as well as

potential discretionary bonuses and the ability to purchase additional stock at a discount through an Employee Stock Purchase Plan. You will also receive access to comprehensive medical, vision, and dental coverage, access to a 401(k) retirement plan, short and long-term disability insurance, life insurance, paid parental leave, and various other discounts and perks. You may also accrue 3 weeks of paid vacation and will be eligible for 10 or more paid holidays per year. Employees accrue paid sick leave pursuant to Company policy which satisfies or exceeds the accrual, carryover, and use requirements of the law.

ITAR REQUIREMENTS:

- To conform to U.S. Government export regulations, applicant must be a (i) U.S. citizen or national, (ii) U.S. lawful, permanent resident (aka green card holder), (iii) Refugee under 8 U.S.C. § 1157, or (iv) Asylee under 8 U.S.C. § 1158, or be eligible to obtain the required authorizations from the U.S. Department of State. Learn more about the ITAR [here](#).

SpaceX is an Equal Opportunity Employer; employment with SpaceX is governed on the basis of merit, competence and qualifications and will not be influenced in any manner by race, color, religion, gender, national origin/ethnicity, veteran status, disability status, age, sexual orientation, gender identity, marital status, mental or physical disability or any other legally protected status.

Applicants wishing to view a copy of SpaceX's Affirmative Action Plan for veterans and individuals with disabilities, or applicants requiring reasonable accommodation to the application/interview process should reach out to EEOCompliance@spacex.com.

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Preferred First Name

Email *

Country *



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Location (City) *

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Enter manually

Accepted file types: pdf, doc, docx, txt, rtf

Education

School *

Select...



Degree *

Select...



Discipline

Select...



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LinkedIn Profile

Portfolio or Cover Letter

[Attach](#)

Accepted file types: pdf, doc, docx, txt, rtf

Attach additional content you'd like us to review with your application. Please combine multiple attachments into a single .zip file.

How did you hear about this job? *

Select...

If event or other, please specify below.

Please specify

GPA (Undergraduate) *

Select...

Please convert your GPA to a 4.0 scale. Select "Not applicable/Do not recall" if you do not have or do not remember your undergraduate GPA.

GPA (Graduate) *

Select...

Please convert your GPA to a 4.0 scale. Select "Not applicable/Do not recall" if you do not have or do not remember your graduate GPA.

GPA (Doctorate) *

Select...

Please convert your GPA to a 4.0 scale. Select "Not applicable/Do not recall" if you do not have or do not remember your doctorate GPA.

SAT Score *

Select...

Please indicate your test score on the 1600 or 2400 scale. Select "Did not take" if you have not taken the SAT.

ACT Score *

Select...

Select "Did not take" if you have not taken the ACT.

GRE Score *

Select...

Active Security Clearance(s) *

Select...

SpaceX & SpaceXAI Employment History *

Select...

This includes internships and full-time employment.

Can you perform all of the essential functions of this role with or without reasonable accommodations? *

Select...

Are you legally authorized to work in the United States? *

Select...

Citizenship Status *

Select...

To conform to U.S. Government space technology export regulations, including the International Traffic in Arms Regulations (ITAR) you must be a U.S. citizen, lawful permanent resident of the U.S., protected individual as defined by 8 U.S.C. 1324b(a)(3), or eligible to obtain the required authorizations from the U.S. Department of State. [Learn more about ITAR here.](#)

If (f) Other, please explain:

Voluntary Self-Identification

For government reporting purposes, we ask candidates to respond to the below self-identification survey. Completion of the form is entirely voluntary. Whatever your decision, it will not be considered in the hiring process or thereafter. Any information that you do provide will be recorded and maintained in a confidential file.

As set forth in SpaceX's Equal Employment Opportunity policy, we do not discriminate on the basis of any protected group status under any applicable law.

Gender

Select...

Are you Hispanic/Latino?

Select...

Race & Ethnicity Definitions

If you believe you belong to any of the categories of protected veterans listed below, please indicate by making the appropriate selection. As a government contractor subject to the Vietnam Era Veterans Readjustment Assistance Act (VEVRAA), we request this information in order to measure the effectiveness of the outreach and positive recruitment efforts we undertake pursuant to VEVRAA. Classification of protected categories is as follows:

A "disabled veteran" is one of the following: a veteran of the U.S. military, ground, naval or air service who is entitled to compensation (or who but for the receipt of military retired pay would be entitled to compensation) under laws administered by the Secretary of Veterans Affairs; or a person who was discharged or released from active duty because of a service-connected disability.

A "recently separated veteran" means any veteran during the three-year period beginning on the date of such veteran's discharge or release from active duty in the U.S. military, ground, naval, or air service.

An "active duty wartime or campaign badge veteran" means a veteran who served on active duty in the U.S. military, ground, naval or air service during a war, or in a campaign or expedition for which a campaign badge has been authorized under the laws administered by the Department of Defense.

An "Armed forces service medal veteran" means a veteran who, while serving on active duty in the U.S. military, ground, naval or air service, participated in a United States military operation for which an Armed Forces service medal was awarded pursuant to Executive Order 12985.

Veteran Status

Select...

Voluntary Self-Identification of Disability

Form CC-305

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OMB Control Number 1250-0005

Expires 04/30/2026

Why are you being asked to complete this form?

We are a federal contractor or subcontractor. The law requires us to provide equal employment opportunity to qualified people with disabilities. We have a goal of having at least 7% of our workers as people with disabilities. The law says we must measure our progress towards this goal. To do this, we must ask applicants and employees if they have a disability or have ever had one. People can become disabled, so we need to ask this question at least every five years.

Completing this form is voluntary, and we hope that you will choose to do so. Your answer is confidential. No one who makes hiring decisions will see it. Your decision to complete the form and your answer will not harm you in any way. If you want to learn more about the law or this form, visit the U.S. Department of Labor's Office of Federal Contract Compliance Programs (OFCCP) website at www.dol.gov/ofccp.

How do you know if you have a disability?

A disability is a condition that substantially limits one or more of your "major life activities." If you have or have ever had such a condition, you are a person with a disability. **Disabilities include, but are not limited to:**

- Alcohol or other substance use disorder (not currently using drugs illegally)
- Autoimmune disorder, for example, lupus, fibromyalgia, rheumatoid arthritis, HIV/AIDS
- Blind or low vision
- Cancer (past or present)
- Cardiovascular or heart disease
- Celiac disease
- Cerebral palsy
- Deaf or serious difficulty hearing
- Diabetes
- Disfigurement, for example, disfigurement caused by burns, wounds, accidents, or congenital disorders
- Epilepsy or other seizure disorder
- Gastrointestinal disorders, for example, Crohn's Disease, irritable bowel syndrome
- Intellectual or developmental disability
- Mental health conditions, for example, depression, bipolar disorder, anxiety disorder, schizophrenia, PTSD
- Missing limbs or partially missing limbs
- Mobility impairment, benefiting from the use of a wheelchair, scooter, walker, leg brace(s) and/or other supports
- Nervous system condition, for example, migraine headaches, Parkinson's disease, multiple sclerosis (MS)
- Neurodivergence, for example, attention-deficit/hyperactivity disorder (ADHD), autism spectrum disorder, dyslexia, dyspraxia, other learning disabilities
- Partial or complete paralysis (any cause)
- Pulmonary or respiratory conditions, for example, tuberculosis, asthma, emphysema
- Short stature (dwarfism)
- Traumatic brain injury

Disability Status

Select...

PUBLIC BURDEN STATEMENT: According to the Paperwork Reduction Act of 1995 no persons are required to respond to a collection of information unless such collection displays a valid OMB control number. This survey should take about 5 minutes to complete.

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